



LOST BOYS CYBER

THREAT INTELLIGENCE • MALWARE ANALYSIS • DETECTION ENGINEERING



OpenPLC v3 CISA ICS Advisory Prioritization

CVE / Vulnerability Threat Intelligence Report

Classification	TLP:CLEAR
Published	2026-07-19
Version	1.0
Author	Lost Boys Cyber
Report Type	CVE / Vulnerability Threat Intelligence Report

TLP:CLEAR | Lost Boys Cyber | OpenPLC v3 CISA ICS Advisory Prioritization - CVE / Vulnerability Threat Intelligence Report

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AI Generation: This report was generated with AI assistance and is provided for informational purposes only. All findings, IOCs, detection rules, hunting queries, attribution assessments, and recommendations must be independently reviewed and validated by a qualified security professional before operational use.

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OpenPLC v3 CISA ICS Advisory Prioritization

1. Executive Summary

CISA published an ICS advisory for OpenPLC v3, making programmable-control and lab/industrial automation exposure a near-term validation priority for OT defenders. Public reporting did not provide enough source-backed evidence to claim known exploitation; the risk is exposure management, compensating controls, and telemetry readiness for an open-source PLC runtime that may appear in labs, training ranges, and unmanaged industrial prototypes.

Lost Boys Cyber selected this item for the 2026-07-19 UTC daily intelligence production run. Source selection preferred material from the last 24-72 hours; where corroborating vendor or government context was not newly published, the lookback expanded to seven days and standing vendor/government guidance. This report does not assert exploitation, actor attribution, affected versions, or IOCs beyond reviewed public-source support.

Field	Assessment
Report family	CVE / Vulnerability Threat Intelligence Report
Severity	High
Confidence	Medium
Published / processed	2026-07-19 UTC
CVEs	No CVE confirmed in selected public sources
Primary defender question	What is exposed, what must be remediated, and what evidence proves mitigation?

2. Vulnerability Metadata

Field	Value
Product / Component	OpenPLC v3 CISA ICS Advisory Prioritization
Weakness	Public advisory details should be checked in the canonical vendor/CISA source before operational use.
Severity	High operational priority
Affected Versions	See primary advisory and vendor documentation.
Fixed Versions	See primary advisory and vendor documentation.
Disclosure / selection date	2026-07-19
Exploitation Status	Not asserted by this report unless listed in CISA KEV or the primary advisory.

3. Source Review & Confidence

Source	URL	Analyst Use
CISA OpenPLC v3 ICS advisory	https://www.cisa.gov/news-events/ics-advisories/icsa-26-190-01	Primary advisory and mitigation source.
CISA ICS advisories index	https://www.cisa.gov/news-events/ics-	Advisory publication and update context.

	advisories	
CISA Control Systems defense-in-depth guidance	https://www.cisa.gov/resources-tools/resources/defense-depth-strategies	Control-system compensating-control guidance.

Confidence is medium because the item is source-backed but some product-version, CVE, or exploitability details require direct validation against the vendor advisory. Defenders should use this report as a prioritization and hunt package, not as a substitute for vendor remediation instructions.

4. Vulnerability Details

The affected technology should be reviewed for internet exposure, privileged administrative paths, default credentials, remote-access dependencies, and integration with sensitive operational workflows. For OT/ICS items, assess whether the product is deployed in production, lab, development, or vendor-managed environments and whether network segmentation prevents direct adversary reachability.

5. Attack Scenarios

- Internet-exposed or partner-reachable systems are probed after public advisory publication.
- A compromised vendor, MSP, or engineer account reaches an internal management console that was assumed to be isolated.
- An attacker uses vulnerability chatter as a phishing pretext against administrators responsible for patching the platform.

6. Threat Landscape & Exploitation Status

No hard IOCs were available in the selected public sources. Prioritize known-exploited status checks, advisory updates, exploit-publication monitoring, and validation of compensating controls. If CISA KEV status changes, update patch SLAs immediately.

7. Affected Products and Exposure

Exposure Pattern	Why It Matters	Immediate Validation
Internet-facing service	Shortens attacker path from advisory publication to probing.	External asset scan and WAF/proxy log review.
Internal privileged console	Compromise can still occur through VPN, help desk, vendor, or admin identity abuse.	Verify MFA, admin group membership, source networks, and session logs.
OT/ICS adjacent deployment	Safety and availability impacts may exceed normal enterprise IT risk.	Confirm segmentation, backup/restore, change windows, and vendor escalation.
Patched or mitigated asset	Evidence is needed for audit and incident triage.	Preserve patch records and successful mitigation screenshots/logs.

8. Mitigation and Workarounds

Apply vendor guidance, restrict management exposure, validate least privilege, confirm backups, and document compensating controls. For high-value systems, add temporary monitoring for advisory keywords, scanning bursts, authentication failures, and unusual administration after patching.

9. Detection Engineering

Signal	Telemetry Source	Analytic Goal
Advisory keyword pivots: openplc, ics, ot, cisa-ics, daily-intelligence	SIEM, EDR, proxy, case notes	Find internal references, scanning, and suspicious command-line research.

Increased 4xx/5xx or admin-path requests	Web, proxy, WAF	Detect post-advisory probing.
New administrator sessions from unusual source networks	IAM, VPN, EDR	Identify identity-mediated exploitation or response activity.
Scripted download or command execution from admin hosts	EDR process telemetry	Surface post-access activity.

```

title: Lost Boys Cyber - OpenPLC v3 CISA ICS Advisory Prioritization
id: lbc-openplc-v3-cisa-ics-advisory-prioritization
status: experimental
description: Behavioral detection scaffold for advisory-driven probing or administrative abuse related to OpenPLC v3 CISA ICS Advisory Prioritization.
logsource:
  category: webserver
detection:
  selection_keywords:
    cs-uri-query|contains:
      - "openplc"
      - "ics"
      - "ot"
      - "cisa-ics"
      - "daily-intelligence"
  selection_admin:
    cs-uri-stem|contains:
      - '/admin'
      - '/login'
      - '/api'
  condition: selection_keywords or selection_admin
falsepositives:
  - Vulnerability scanners, patch validation, or administrator testing
level: medium

```

10. Threat Hunting

Hypothesis: exposed or privileged systems show probing, failed authentication, or unusual administrator activity after public advisory attention.

```

DeviceNetworkEvents
| where Timestamp > ago(14d)
| where RemoteUrl has_any ('openplc', 'ics', 'ot', 'cisa-ics', 'daily-intelligence') or
InitiatingProcessCommandLine has_any ('openplc', 'ics', 'ot', 'cisa-ics', 'daily-intelligence')
| summarize FirstSeen=min(Timestamp), LastSeen=max(Timestamp), Connections=count(),
Samples=make_set(RemoteUrl, 20) by DeviceName, InitiatingProcessAccountName, InitiatingProcessFileName
| order by LastSeen desc

SigninLogs
| where TimeGenerated > ago(14d)
| where ResultType != 0 or ConditionalAccessStatus in ('failure','notApplied')
| summarize Attempts=count(), Apps=make_set(AppDisplayName, 20), IPs=dcount(IPAddress) by
UserPrincipalName, bin(TimeGenerated, 1d)
| where Attempts > 10 or IPs > 3

```

11. MITRE ATT&CK Mapping

Technique	Name	Evidence / Rationale
T1190	Exploit Public-Facing Application	Relevant when affected systems are exposed to untrusted networks.
T1078	Valid Accounts	Common path for administrator-console and vendor-access abuse.
T1059	Command and Scripting Interpreter	Relevant for post-access validation and

		payload execution.
T1046	Network Service Discovery	Plausible after initial access to locate adjacent assets.

12. Validation / Lab Testing

No exploit reproduction was attempted. Validation completed: source triage, exposure-analysis drafting, detection/hunt scaffolding, PDF render, text extraction, and publication/upload workflow checks.

13. Recommended Actions Summary

Priority	Owner	Action
0-24 hours	Asset / VM	Identify affected products and compare against vendor fixed versions.
0-24 hours	Network / OT	Restrict management interfaces and verify segmentation.
24-72 hours	SOC	Run included hunts and review WAF/proxy/admin logs.
24-72 hours	IR	Prepare evidence package for any confirmed exposure or failed mitigation.

14. References

- [1] CISA OpenPLC v3 ICS advisory: <https://www.cisa.gov/news-events/ics-advisories/icsa-26-190-01>
- [2] CISA ICS advisories index: <https://www.cisa.gov/news-events/ics-advisories>
- [3] CISA Control Systems defense-in-depth guidance: <https://www.cisa.gov/resources-tools/resources/defense-depth-strategies>